Soil wetting state and preferential transport of *Escherichia coli* in clay soils

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Tallon, L. K., Si, B. C., Korber, D. and Guo, X. 2007. **Soil wetting state and preferential transport of *Escherichia coli* in clay soils**. Can. J. Soil Sci. **87**: 61–72. Transport of *Escherichia coli* (*E. coli*) through soil to drinking and recreational water may pose a serious health risk. The objective of this study was to determine how initial preferred soil wetting state influences the preferential transport of *E. coli* in a clay soil. A strain of *E. coli* marked with green fluorescent protein (gfp) was gravity-fed-sprinkler-applied as simulated rainfall to three replicates in different wetting states, along with Cl^- and adsorptive dye near Plenty, SK. Canada. After 48 h, a 50 × 50 × 50 cm^3 block was excavated to determine the transport pathways. Digital image analysis of horizontal sections provided estimates of dye coverage. *Escherichia coli* were significantly filtered in the top 10 cm of soil with concentration profiles similar to that of Cl^-. Ratios of *E. coli* to Cl^- did not show significant differences among treatments ($P < 0.05$) and indicated that below 10 cm depth, *E. coli* and Cl^- were preferentially transported along the same pathways with no significant difference between plots. Results show that the majority of *E. coli* and Cl^- were filtered when transported through the discontinuous pores of the near-surface matrix and suggest a saturated layer that controlled infiltration into organized root channels, resulting in preferential flow.

Key words: Preferential flow, *Escherichia coli*, wetting state, Vertisol, conducting areas

Tallon, L. K., Si, B. C., Korber, D. et Guo, X. 2007. **Humidification du sol et mode de transport d'élection de *Escherichia coli* dans les sols argileux**. Can. J. Soil Sci. **87**: 61–72. Le transport du colibacille (*Escherichia coli*) dans le sol jusqu'aux réserves d'eau potable et aux étendues d'eau récréatives pourrait poser un risque sérieux pour la santé. La présente étude devait préciser le rôle que l'humidification initiale du sol joue dans le transport que le colibacille privilégie dans les sols argileux. À cette fin, les auteurs ont arrosé trois parcelles avec une souche du colibacille marquée au moyen d'une protéine fluorescente verte par aspersion pour imiter la pluie et obtenir trois degrés d'humidification près de Plenty (Saskatchewan), au Canada. Les arrosages incluaient des ions Cl^- et un colorant adsorbant. Quarante-huit heures plus tard, ils ont excavé un bloc de 50 × 50 × 50 cm^3 de sol afin de déterminer le trajet suivi. L'analyse visuelle numérique des tranches horizontales a permis d'estimer la distance couverte par le colorant. Le colibacille traverse sensiblement les 10 premiers centimètres du sol, sa concentration se rapprochant de celle des ions Cl^-. Le ratio entre la concentration de colibacilles et celle d'ions Cl^- ne varie pas significativement entre les traitements ($P < 0,05$), signe qu'à plus de 10 cm de profondeur, colibacilles et ions privilégient les mêmes modes de transport sans différence notable d'une parcelle à l'autre. Ces résultats indiquent que le passage à travers les pores discontinus de la matrice proche de la surface filtre la majeure partie des colibacilles et des ions Cl^-. Une couche saturée pourrait donc réguler l'infiltration à travers un réseau de canaux radiculaires, ce qui engendrerait la canalisation de l'écoulement.

Mots clés: *Escherichia coli*, humification initiale du sol, Vertisol

Intensive beef feedlots are becoming increasingly prevalent on the semi-arid prairies of North America. With a rise in the number of cattle comes an increase in the amount of manure to be disposed of. While cattle wastes are typically land applied as a soil amendment, the available area on which wastes are spread cannot increase concomitantly. Cattle manure is the primary reservoir of *Escherichia coli* 0157:H7 (*E. coli*) in the environment (Jones 1999). Thus, spreading manure on areas in close proximity to susceptible water resources raises serious health concerns. Fecal coliforms can include pathogens such as *Escherichia coli* 0157:H7; a virulent strain that is infectious at numbers as low as 10 cells (Paton and Paton 1998). For infectious outbreaks to be avoided, the transport pathways of *E. coli* and other fecal coliforms to water sources must be examined. Overland transport of fecal coliforms can be minimized effectively by incorporation of manure into the soil; however, transport through the soil profile to groundwater still presents a serious concern.

Bacteria are transported through the soil via flowing water. However, given their size and surface properties, bacteria are subject to transport attenuation. There are myriad ways in which bacteria can be removed from the convective solution. Retention mechanisms can include filtration, where bacteria are trapped in pores that are too small to allow transport, sorption, where bacteria attach to particle surfaces despite being transported in pores of greater size than the bacteria, and capture at the air-water interface (AWI), where surface tension and electrostatic forces trap bacteria at the interface of soil water and insular air bubbles (McMurray et al. 1998). Retention of bacteria is maximum when matrix flow dominates transport as water flows through micropores, especially in fine-textured soils at low velocities (Abu-Ashour et al. 1998; Unc and Goss 2003;

Aislabie et al. 2001; Unc and Goss 2003; Abu-Ashour et al. 1998). Conversely, the presence of macropores, which, for the purposes of this manuscript, refer to pores that empty completely at ≤ 3 cm tension corresponding to an equivalent pore diameter of 1 mm, along with soil heterogeneity, can lead to enhanced transport resulting in bacteria bypassing the bulk soil matrix (Smith et al. 1985; Natsch et al. 1996; Aislabie et al. 2001; Unc and Goss 2003).

Heterogeneous and inconsistent wetting of soil, termed preferential flow, can arise due to heterogeneity in pore size distribution. Preferential flow leads to inconsistencies in the infiltration of water into soil. This is important since many processes, such as infiltration, heat fluxes, contaminant transport, hydraulic conductivity, and matric potential will vary with soil water content in a non-linear fashion (Western et al. 2002). Soils with high antecedent water content have high matric potentials (because matric potential is negative), high hydraulic conductivities, and a greater number of hydraulically active large pores that conduct more water at a faster rate. Conversely, resistance to flow increases rapidly at low soil water contents (Western et al. 2002). This implies that at low water contents only small pores are filled and conduct water at a slower rate, according to capillary theory. The non-linear relationship that transport processes can have with soil water content led Grayson et al. (1997) to propose the idea of preferred wetting states for soil. Grayson et al. (1997) theorized that different processes dominate redistribution of water through soils in different preferred states whether the soil is wet, dry, or in the process of actively wetting. Although there are no prior accounts in the literature, this idea may be applied to contaminant transport. The initial soil preferred state would therefore affect which transport processes are active, and influence the preferred soil transport mechanisms. Throughout the manuscript, the initial soil wetting state influencing the preferred soil transport mechanisms will be referred to as the preferred state.

Given that transport processes are modified with changes in soil water content, it is reasonable that initial soil wetting state can lead to different dominant processes and dictate how bacteria are transported. In a dry state, surface runoff into desiccation cracks can be very high, but if cracks are not present the low soil matric potential could also lead to an even distribution of water in the bulk soil matrix. In a wet soil, despite high water content, macropores by definition will be drained. Filtration could be high given that bacteria are able to sample a larger soil volume and are thus exposed to a greater number of potential attachment sites, although macropores would be primed and readily able to conduct water. Under heavy rainfall (soil is close to saturation) soil macropores are filled, thus dominating transport.

Typically, conditions conducive to preferential flow events such as near-saturated macropores only exist in the spring after snowmelt and runoff in semi-arid and arid zones. In semi-arid agricultural areas these conditions rarely arise. The growing season climate is generally characterized by evapotranspiration exceeding precipitation and a dry preferred state (Grayson et al. 1997). Periods of drought are periodically interrupted by intense convective thunderstorms that cause a switch to a wet preferred state, organizing soil water into consistent, connected flow paths (Grayson et al. 1997).

Dominant trends have been reported on the prevailing soil conditions leading to the preferential transport of bacteria and chemicals. Conditions can include a sufficient number of macropores connected to the surface, high initial water content, high infiltrating water flux density, the structure of the soil, and interaction of low soil permeability and high initial water content (Zehe and Flühler 2001; Weiler and Naef 2003; Flury and Wai 2003). Kung et al. (2000a, b) found that preferential flow will dominate chemical transport in a wet soil regardless of the chemical's retardation properties. Microbial transport studies often state that if macropores exist and there is sufficient water for saturation, movement of bacteria through the soil profile will be dominated by preferential transport (Natsch et al. 1996; McMurry et al. 1998; Aislabie et al. 2001).

Though receiving little attention, desiccation cracks are potentially important preferential transport pathways for bacteria. When clay soils are subject to wetting and drying cycles, they are prone to swelling and cracking. It has been noted that cracks can be important in the transport of chemical contaminants to depth (Velde et al. 1996; Nobles et al. 2003; Artz et al. 2005). Transport along desiccation cracks can be the dominant mechanism for deep wetting of vertisols and will have a direct influence on solute pathways. When desiccation cracks have received sufficient water, they will swell and close, effectively sealing a large transport pathway. Despite sealed cracks, for soils in a wet state the majority of transport can be along root channels and other macropores (Nobles et al. 2003). Root channels and the distribution of other macropores in a clay soil can lead to a partitioning of transport systems. These mechanisms are well known under steady-state conditions. However, transport mechanisms for transient conditions in semi-arid areas, where rainfall distribution is not uniform and occurs as infrequent, intense storms, are not well known. Owing to the paucity of data in the literature it is not clear which preferred state is more conducive to initiating preferential transport. Nor is it known whether desiccation cracks or root channels are more dominant in preferentially conducting bacteria through the soil in semi-arid areas. This is an area of research that needs to be addressed.

The transport of *E. coli* and faecal coliforms through soil has been extensively studied both in the field (Natsch et al. 1996; McMurray et al. 1998; Aislabie et al. 2001; Celico et al. 2004) and in the lab (Smith et al. 1985; Abu-Ashour et al. 1998; Gagliardi and Karns 2000; Saini et al. 2003; Artz et al. 2005). However, there are very few data on the dominant transport mechanisms or on how bacteria infiltrating into a soil of a particular preferred state (wet, dry, wetting) are transported. This is especially true for the transport of *E. coli* through clay soils displaying vertisolic properties. Therefore, the objectives of this study are twofold: (1) to examine the dominant *E. coli* transport mechanisms as a function of depth; and (2) to compare and contrast the transport mechanisms in relation to antecedent soil preferred wetting state. The results of this study will aid in the understanding of the mechanisms by which *E. coli* are transported in clay soils.

MATERIALS AND METHODS

Site Characteristics

This study was conducted on the semi-arid prairies at a site near Plenty, Saskatchewan, Canada. Annual precipitation and average annual air temperature for this site are 327 mm and 3°C, respectively. The site was situated in the Dark Brown Soil Zone and the soil was classified as a Vertic Brown Chernozem soil of the Regina Association and ranges from clay loam to clay texture. Soil physicochemical properties are given in Table 1 (Zeleke 2003). The soil hydraulic properties (saturated hydraulic conductivity, inverse capillary length scale, and macroporosity) were determined using a disc infiltrometer (Soil Measurement Systems, Tucson, Arizona). Chemical properties are only given for the 0- to 10-cm depth since the absence of horizonation made the soil uniform over the studied depth. The site was managed under a cereal/ fallow rotation and in 2004 was cropped to canary seed. The experimental plots were harvested by hand using clippers before the experiment.

Experimental Design

The study examined the effect of soil wetting state on the transport of *E. coli* through the soil. *Escherichia coli* and a suite of tracers were applied to soil in different wetting states at the time of *E. coli* application. The treatments were organized as follows: (1) Dry state, to examine a soil with low matric potential and where water in the pores is highly disconnected; (2) Actively Wetting state, to examine a soil with hydraulically active channels (water is flowing in soil profile); and (3) Wet state, to examine a soil with minimum flow, but high initial water content (at field capacity). Cracks on the surface are still wide open in the Dry state, somewhat open for the Actively Wetting state, and likely closed in Wet state. These three states simulate scenarios where (1) heavy rainfall falls on dry soil surface with manure (Dry state); (2) grazing during a prolonged period of rainfall (Actively wetting state); and (3) rainfall falls on soil surface where manure was applied just before snow melt (Wet state). Initial and final water contents were measured immediately before and after tracer application using vertically oriented time domain reflectometry (TDR) probes of 10 cm long (Tektronix, OR). The initial water contents were taken at locations about 5 cm away from a crack. Most cracks were closed (but still visible) after water applications due to swelling of clay particles and the final water contents were taken near the closed cracks. Travel distance readings from TDR were substituted into the empirical relationship given by Topp et al. (1980) to arrive at volumetric water content. Experimental plots had an area of 100×100 cm^2. Three treatments were arranged in a randomized complete block design and replicated three times.

Two tracers, Chloride (Cl^-) and Brilliant Blue FCF dye (Keystone Aniline, IL) along with *E. coli* were used in this experiment. Chloride was used as a conservative tracer to determine the flow paths of the input water and chemicals that do not interact with the surrounding soil. Chloride was applied as KCl (Panther Chemicals, Canada) at an input molarity of 0.03 M coinciding with the estimated molarity of beef manure (Unc and Goss 2004). To visually determine the water flow pathways, Brilliant Blue FCF dye (Colour Index 42090) was added as an adsorptive tracer at a rate of 4 g L^{-1}. Tracers were applied with a gravity -fed sprinkler as part of a 1.9 cm simulated rainfall solution.

Escherichia coli Preparation

A non-pathogenic *Escherichia coli* isolate from a meat packing plant in Lacombe, Alberta, Canada was used in this experiment. The *E. coli* isolate, known as Strain 8 was genetically modified with a mini-Tn5 transposon, carrying the gene for green fluorescent protein (gfp) and is resistant to the antibiotic kanamycin (Kn) (Suarez et al. 1997). Strain 8 was cultured on 10 % trypticase soy agar (TSA-Kn) (Becton Dickinson, MD) containing 50 μg mL^{-1} kanamycin acid sulfate.

Cells used in the experiment were harvested in the middle of the exponential growth phase. One mL of an overnight broth was used to inoculate 50 mL of trypticase soy broth (TSB) (Becton Dickinson, MD). After 3.75 h, the cells were separated from the media by centrifugation, washed twice with sterile physiological saline, re-suspended in 10 mL of saline and transferred to 90 mL of 0.1 M saline to arrive at an input solution volume of 100 mL. This was added to 900 mL of distilled water to later become the *E. coli* input solution. The 1 L *E. coli* input solution had a concentration of 10^6 *E. coli* cfu mL^{-1} (10^9 cfu L^{-1}), which coincides with the approximate concentration of faecal coliforms in beef manure (Unc and Goss 2004). Assuming a typical application rate of 2×10^4 kg cattle manure ha^{-1}, the input rate in this experiment corresponds to an input density of 8×10^8 cfu m^{-2} to 2×10^9 cfu m^{-2} and is what would typically be applied to soil (Unc and Goss 2004).

Treatment Application

The objective of the study was to examine the influence of preferred soil state (soil water states) on *E. coli* transport. Thus, the experiment began with each treatment in different preferred states. For the wet state treatment, 4 cm of water were applied over the entire 10 000 cm^2 to obtain a soil of higher soil water content (θ_v) than the dry state and actively wetting state treatments, which did not receive a pre-treatment wetting. The 4 cm of water applied was allowed to redistribute for 48 h. The average final 0 to 10 cm θ_v of the wet state treatment after application of pre-treatment wetting was 0.59 ± 0.04 cm^3 cm^{-3}.

Forty-eight hours after the initial water application, the experiment began with the application of *E. coli* and tracers to the dry state, actively wetting state, and wet state treatments. Since the actively wetting state treatment examined the influence of a soil with hydraulically active transport pathways, *E. coli* had to be applied during wetting of the soil, or, in this case, mid-way through application of tracers. The sequence of applications was as follows: (1) pre-wet application for wet state, (2) redistribute 48 h, (3) application of *E. coli* on dry and wet state treatments, (4) immedi-

Table 1. Physicochemical properties of a clay loam soil measured at various depths for Plenty, SK. Measured parameters include bulk density (ρ_b), electrical conductivity (EC), total carbon (C), inverse capillary length scale (α), and saturated hydraulic conductivity (K_s) (Zeleke 2003)

Depth cm	Sand	Silt (%)	Clay	ρ_b ($Mg\ m^{-3}$)	pH	EC ($dS\ m^{-1}$)	Total C ($Mg\ ha^{-1}$)	α (m^{-1})	K_s ($m\ s^{-1}$)	Macorporosity[z] ($m^3\ m^{-3}$)
0–10 (plough layer)	20.8	44.3	34.9	0.5	8.3	0.3	25.8	23	3.01×10^{-5}	0.073
10–30	14.8	32.9	52.3	1.24						
30–60	14	32.1	53.9	1.34						

[z] Pores ≥ 1 mm in size as determined at 3 cm tension.

ate application of Cl^- and dye as simulated rainfall to all treatments, (5) midway through rainfall *E. coli* is applied to actively wetting state treatment, (6) apply remainder of tracers. Initial and final water contents for all three treatments are given in Table 2. The measured soil water storages were 37 to 84% higher than the water applied, further suggesting that water was preferentially redistributed into cracks, resulting in high water content near cracks. Input subsamples were taken during application for determination of initial concentrations of both bacteria and water-soluble tracers.

To apply *E. coli*, the 100 mL input solution of *E. coli* was added to 900 mL of distilled water to arrive at a 1 L per plot input solution. The 1 L input solution was then sprayed over an inner area of $50 \times 50\ cm^2$ to avoid the edge effects of the sample area. The 1 L input solution applied over 2500 cm^2 resulted in an additional 0.4 cm to be added to the final input solution. The input solution containing Cl^- and dye was applied uniformly by steadily moving a hand-held spray nozzle over the total 10 000 cm^2 area as part of the simulated rainfall. In total, the input tracer solution included 15 L of distilled water and dissolved tracers, plus 1 L of *E. coli* solution, and was applied at a rate of 22.5 cm h^{-1} corresponding to a 25-year return period storm (Gray 1973) to arrive at a total rainfall of 1.9 cm.

Sampling

Tracers and *E. coli* were allowed to redistribute through the soil profile for 48 h. After 48 h the plots were excavated to determine the final distribution of *E. coli* and the tracers. Soil surrounding the inner plot was excavated down to a depth of 50 cm, leaving an inner block measuring $50 \times 50 \times 50\ cm^3$. For some plots the dry soil surrounding the wetted area made excavation by hand impossible. Therefore, these plots were only excavated down to 40 cm, but all replicates had at least one plot excavated to 50 cm. A $50 \times 50\ cm^2$ grid with 10 -cm increments on the x and y geometric axes was laid over the soil surface to demarcate 25 grid sections. Stained areas within each $10 \times 10\ cm^2$ grid location were sampled along with one to two unstained locations for each depth increment (10 cm). One sample per grid section, weighing 30 to 50 g was taken by using a flame-sterilized trowel. The samples were then placed in a sterile 120-mL sample jar and were transferred to a cooler. At the end of the day the coolers were placed in a 4ºC fridge until analysed within 3 wk. Once a given depth was sampled, the next 10 cm was carefully excavated using hand trowels by lifting away thin vertical slices. Care was taken to avoid cross-contamination by reducing contact between stained and unstained areas. The newly exposed surface was then cleared by lightly sweeping and blowing away any loose soil particles. Once the surface was prepared, samples were collected in the same manner as above.

Digital Analysis of Stained Areas

Prior to any excavation of samples, the dye coverage area of the soil was determined and compared with the un-stained area. A flat surface was prepared to allow good contrast between the stained and un-stained areas, and the plot area was shaded from direct sunlight. One photograph for each depth was taken using a digital camera positioned at a consistent height above the profile for all depths. Photographs were downloaded to a PC and analyzed using PCI Geomatica v. 9.1 (PCI Geomatics, Canada). The program was trained manually to separate stained areas from those that were not stained. After the operator selected representative stained areas, the program quantified the total percentage of pixels that were within that colour range. The total area of stained soil (cm^2) was divided by the total sample area of 2500 cm^2 for determination of percent area coverage. For more accurate determination of preferential pathways, other methods such as Weiler and Fluehler (2004) and Kulli et al. (2003) may be adopted. However, the dye patterns were used here only to show the percentage of area responsible for the majority of transport in the system.

Microbial Sample Analysis

For the determination of *E. coli* concentration, a 5 g subsample of soil was weighed out into a square dilution bottle containing 95 mL of sterile physiological saline and was vigorously shaken by hand for 1 min. Aliquots of 1, 0.1, and 0.01 mL were transferred to an empty sterile Petri dish. TSA-Kn tempered at 52˚C was then poured into the plates and allowed to set. Plates were incubated at 37ºC for 36 to 48 h until *E. coli* colonies were of a differentiable size, and could be identified based on fluorescence using a hand-held ultra violet light. Soil subsamples were then air dried at 105°C for 12 h, ground, and water content measured gravimetrically. Plate counts were adjusted for the water content and are reported as colony forming units per gram of dry soil (cfu g^{-1}).

Positive and negative control experiments were conducted on both inoculated and non-inoculated controls. For the inoculated controls, a 10-g sterile soil sample was placed into a 50-mL centrifuge tube and inoculated with 1 mL of an *E. coli* suspension. The tube was vortexed and immediately sampled following the aforementioned procedure.

Table 2. Summary of initial and final volumetric water contents for Dry State, Actively Wetting State, and Wet State treatments for 0- to 10-cm depth. Initial and final water contents are before and after application of *E. coli*, respectively

	Dry		Actively Wetting		Wet	
Treatment	Mean	SD	Mean	SD	Mean	SD
			($cm^3\ cm^{-3}$)			
$\theta_{v\ initial}$	0.19	0.03	0.25	0.03	0.33	0.03
$\theta_{v\ final}$	0.54	0.08	0.56	0.04	0.59	0.04
$\Delta\ \theta_v$	0.35	0.07	0.31	0.07	0.26	0.07
WSI[z] (cm)	3.5		3.1		2.6	

[z]WSI (soil water storage increase) was calculated as $\Delta\ \theta_v$ multiplied by measurement depth.

Chloride Analysis

Soil Cl^- was extracted via filtration using a 2:1 water:soil (mL:g) method. Chloride was indirectly measured using a SpectrAA 220 atomic absorption spectrometer (Varian, Canada). A standard amount of $AgNO_3$ was added to precipitate Cl^- in solution. The remaining Ag that had not precipitated was then measured using a Silver lamp emitting at 328.1 nm (Varian 1985). The method was accurate to within ± 5 mg L^{-1}. Background levels of soil Cl^- were subtracted for final determination of soil Cl^- concentration.

Statistics and Estimation of Mass Recovery

Profile concentration data obtained from horizontal sections for *E. coli*, Cl^- and dye coverage area approximated a logarithmic normal distribution. Therefore, the data were log-transformed using natural logarithms. Analysis of variance (ANOVA) and significance test were performed on the log-transformed concentration data. Least-squares means tests of significance were carried out at the 5% level ($P = 0.05$). However, arithmetic means and standard deviations are given in figures and text, because mass is only conserved with arithmetic means and it is easier to understand mass recovery using mass based measure than using log(mass) based measure. For the estimation of the cumulative infiltration with respect to each 10 cm depth increment, the percent area of soil stained with dye was multiplied by the depth increment, dry bulk density, and the average value of concentration of *E. coli* and Cl^-. The total cfu (*E. coli*) and mass (Cl^-) recovered were then compared with the total cfu and mass applied for the final estimation of cumulative infiltration and recovery and are given as a percentage. Cumulative infiltration was then analyzed using repeated measures ANOVA to characterize trends with respect to depth, treatment and the interaction of the two (Meredith and Stehman 1991). Statistical analyses were carried out using Proc GLM in SAS v.8 (SAS Institute, Inc., Cary, NC). Ratios reported in this study are the ratio of the concentration of *E. coli* and Cl^- at a given increment, normalized to input *E. coli*:

Cl^- ratio $([C_{E.\ coli}/C_{Cl^-}]/[C_{0\ E.\ coli}/C_{0\ Cl^-}])$.

RESULTS AND DISCUSSION

Quantification of Conducting Area

Digital analysis of stained regions of soil indicated that only small areas of soil relative to the total application area, were conducting the applied solution to depth (Figs. 1 and 2). This suggested that preferential flow occurred in all three treatments. The percentage of area conducting water through the profiles was similar irrespective of wetting state, and there were no significant differences between treatments. Any water that bypassed the surface was organized into concentrated flow paths and was preferentially transported to greater depths than would be expected if there were an even distribution of the solution. The dye coverage area given in Fig. 2 is preliminary evidence that all three treatments conducted solution as in a wet state, regardless of antecedent soil water content.

Depth Distribution of *E. coli* and Tracers

Control experiments indicated that gfp-labelled *E. coli* were not present in the soil, and background levels of soil Cl^- detected in the un-stained areas were 14 mg g^{-1}. Therefore, the un-stained data are not reported. Control experiments revealed an *E. coli* recovery of 10%.

The concentration of *E. coli* recovered from stained areas in the experimental soil was small in comparison to the amount of *E. coli* that was applied; the concentration of *E. coli* relative to its input concentration (C/C_0) was 0.1 and 0.09% at the soil surface and declined to less than 0.005 and 0.001% at 50 cm for the dry and wet state treatments, respectively (Fig. 3A and C). Even smaller concentrations were found in the actively wetting state treatment, where C/C_0 was never greater than 0.02% (Fig. 3B). Within treatments, there was a statistically significant accumulation of *E. coli* and Cl^- at the surface increment, whereas increments deeper than 10 cm were not statistically different from each other. Reports in the literature on total recovery range from 64 and 79% (Natsch et al. 1996) to 2.0% (Vinten, et al. 2002). Total recoveries of *E. coli* in the present study were even lower. Selective media and intense fluorescence of the gfp *E. coli* made the bacteria easy to detect in low numbers, suggesting that enumeration method does not account for the low recovery. This would indicate losses in the field. Although reports on the survival of *E. coli* in soil vary widely from a half life of 1.8 to 2.9 d in the field (Vinten et al. 2002) to 41 d in fallow soil to more than 500 d in frozen soil (Gagliardi and Karns 2000), the low recovery was most likely a function of both a strain that is unfit to compete in this environment and low input numbers (Unc and Goss 2004; Topp et al. 2003). It has been found that when soil is inoculated with environmental and enteric strains, the enteric strain will only grow when conditions are favourable and will compete poorly with a soil isolate (Topp et al. 2003).

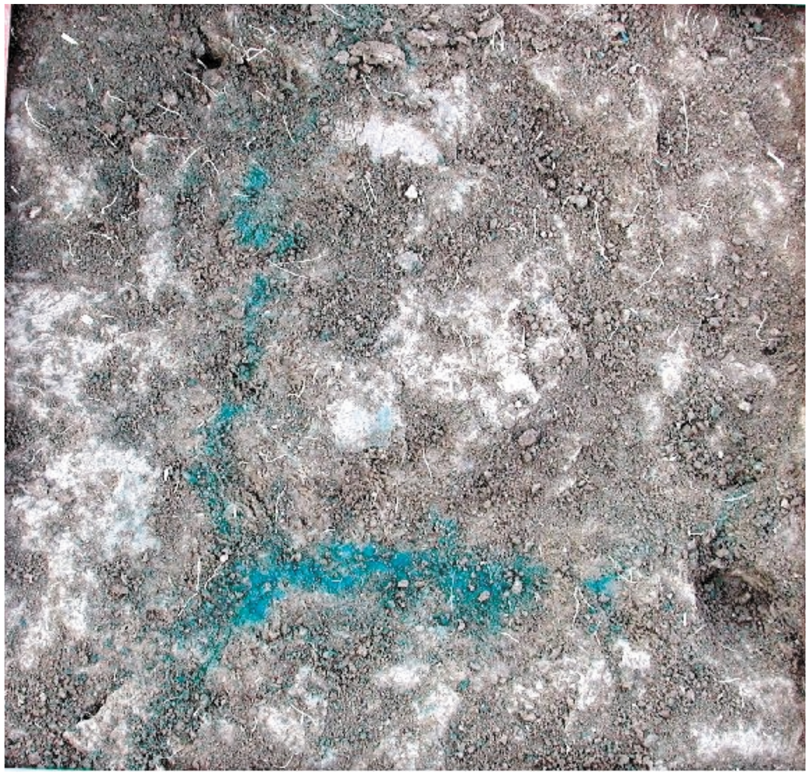

Fig. 1. An example of pictures taken after excavation for determination of area of conductance. A square area was excavated to a depth of 50 cm and was photographed. The picture shows the dye patterns in the plane that parallels the soil surface. Calculation through PCI Geomatics showed the dye coverage is about 2.39% of total area.

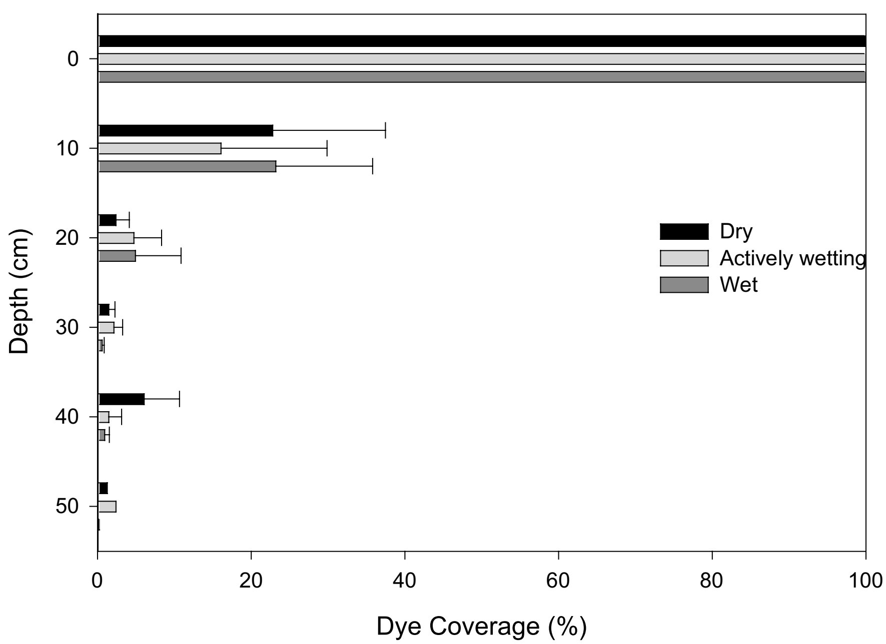

Fig. 2. Mean percent dye coverage for the horizontal cross sections of Dry State, Actively Wetting State, and Wet State treatments. The measurements were taken at discrete depths, and 0 indicates a surface measurement.

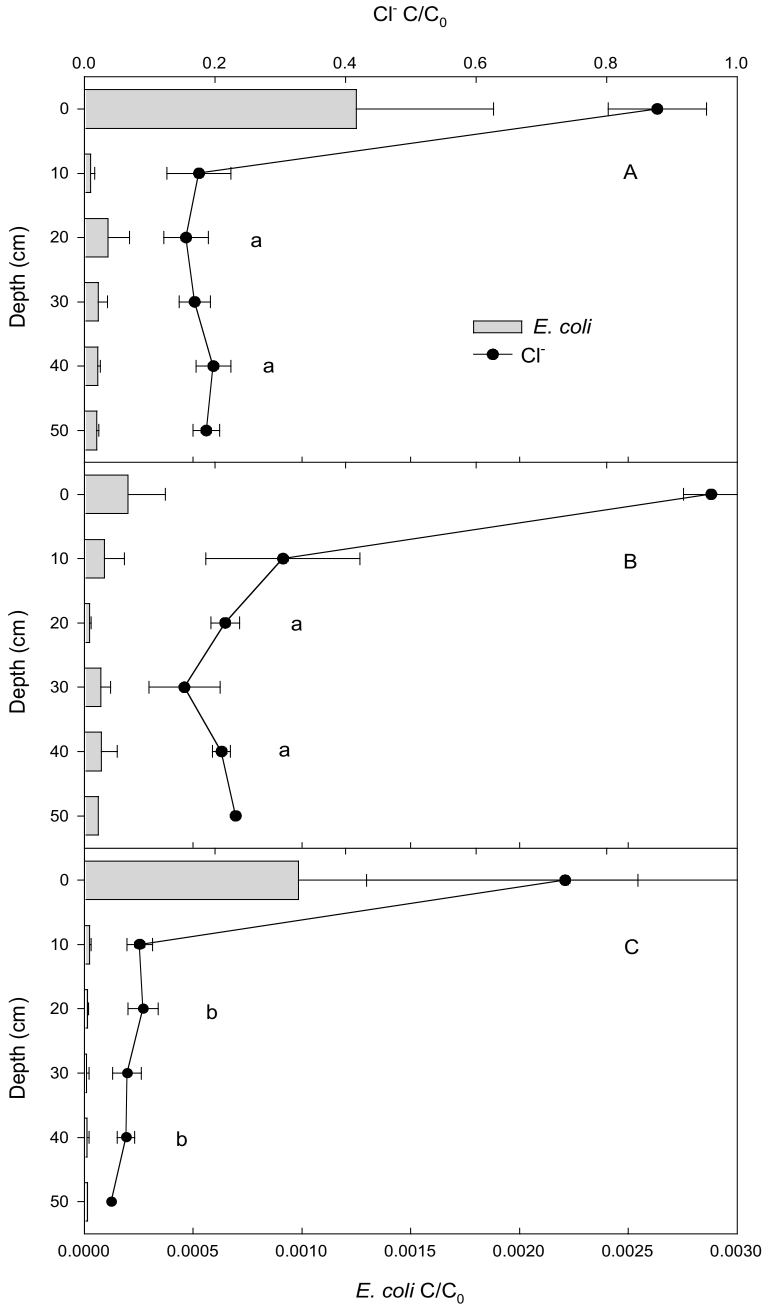

Fig. 3. Depth distribution of *E. coli* and Cl^- concentrations (C) in stained areas of soil relative to input concentrations (C_0) for Dry State (A), Actively Wetting State (B), and Wet State (C) treatments. Lower case letters indicate statistically significant differences between treatments at $P < 0.05$. The measurements were taken at discrete depths, and 0 indicates a surface measurement.

There were no statistically significant differences between wetting state treatments in the concentration distribution of *E. coli*. What is of note, however, is the corresponding trend in C/C_0 of Cl^- with respect to *E. coli* (Fig. 3A, B, and C). In all treatments, it can be seen that Cl^- was deposited mainly in the upper 10 cm of the profile, with a low level of breakthrough further down. This deposition trend corresponds closely to the one exhibited by *E. coli*. Below z = 10 cm, only the wet state treatment had a significantly different mean from the dry and wetting state treatments and even then only in the 10 to 20 cm and 30 to 40 cm depths. The similar transport results across all treatments are evidence for the organization of flow paths below the surface of the soil, a diagnostic factor of the wet state proposed by Grayson et al. (1997).

Filtration, sorption, or AWI capture could be responsible for the large decrease in concentration in the 0- to 10-cm increment (Fig. 3). However, given the fine-textured soil and discontinuous pore network in the 0- to 10-cm plough layer, filtration was most likely predominant. Initial filtration at shallow depths compares well with others. In column experiments, Saini et al. (2003) found that the majority of *E. coli* applied to a fine loam soil were found near the surface. Unc and Goss (2003) predicted potential maximum transport accumulations in sandy loam and silt loam soils to be ~30% near 20-cm depth, with a decrease to near 0 at 40 cm and beyond. In half of the 45 cm undisturbed sandy loam soil cores used by Artz et al. (2005) there was <1% breakthrough, with the rest retained in the soil matrix. An important point to note is that deposition of *E. coli* and Cl^- in similar proportions suggested that despite differences in adsorptive properties, the two were being transported along the same pathways through the soil profile.

Escherichia coli to Cl^- Ratios

Examining the concentration distribution of *E. coli* provides insight into the location of bacteria transport, but not the mechanism of transport. In order to gain an understanding of the processes involved, two tracers with differing adsorption characteristics can be examined (Natsch et al. 1996). To address this, the ratio of adsorptive *E. coli* to conservative Cl^- normalized to their input and their relative changes with depth can be compared (Fig. 4A, B, and C). The initial input ratios of *E. coli* to Cl^- were 469, 493 and 602 cfu g^{-1}, for dry, actively wetting, and wet state treatments, respectively. At the 0- to 10-cm increment the measured ratios indicated a substantial drop in *E. coli* concentration. The general trend in ratio is similar to the concentration distribution of *E. coli*, where the ratio of *E. coli* to Cl^- was initially high in the upper 10 cm and then decreased as depth increased for dry state and wet state treatments (Fig. 4A, B, C). Unlike the concentration distribution, the ratio below the surface increment remained consistent with an increase in depth. The trend of a consistent ratio below 10 cm was similar for all treatments and there were no significant differences among them. The findings of the present study are consistent with those of Natsch et al. (1996) where *P. fluorescens* were filtered in the surface layer, and any transport to depth occurred through organized channels. A consistent ratio with increasing depth indicated that transport mechanisms were not different between treatments and *E. coli* were being transported through the same organized flow paths, which supports the idea that all treatment soils were in a wet preferred state.

Cumulative Infiltration

To analyze the interaction of the three treatments with respect to cumulative infiltration, repeated measures ANOVA were conducted (Tables 3 and 4). To facilitate this, cfu of *E. coli* and mass of Cl^- was used instead of concentrations. Preliminary evidence in Figures. 5 and 6 showed that trend lines did not diverge and followed the same general pattern. The similar trends indicated that there was no statistically significant treatment effect at different depths. Repeated measures analysis as shown in Tables 3 and 4 revealed four key points: there were no significant differences between treatments (Between Treatment); the trend with increasing depth was similar for all treatments (Wilks λ Depth); there was no interaction between depth and treatment (Wilks λ Depth $\times$ Treatment); and all treatments followed a similar trend with respect to depth. Furthermore, the trend with depth is described significantly with a linear equation (Mean Linear). In addition, it is apparent that the variation of *E. coli* and Cl^- for each depth was similar throughout the profile. Indeed, cumulative infiltration patterns of *E. coli* and Cl^- were highly correlated for dry (R^2 = 0.87), actively wetting (R^2 = 0.99) and wet (R^2 = 0.99). Essentially, the cumulative infiltration data point to the conclusion that all three treatments were in the same preferred state, which was a wet state.

There is substantial evidence in this study that *E. coli* and Cl^- were transported in the same manner for all treatments, partially accumulating in the soil matrix in the surface 10 cm increment, and partially by-passing soil matrix and entering organized flow paths in the subsurface. *E. coli* and Cl^- were transported similarly through both of these layers. Significant concentrations of *E. coli* and Cl^- in the upper 10 cm are evidence of the former, consistent *E. coli* to Cl^- ratios is evidence of the latter, and similar dye staining patterns in all treatments is evidence for both. Matrix flow through the plough layer increased filtration and sorption because of the less continuous pore structure and smaller pore diameter. This allows for lower water velocities, smaller channels, and closer proximity to attachment sites. Tillage at the surface will serve to disconnect smaller surface pores from underlying macropores. Areas with a discontinuous pore network characterized by low permeability and low saturated hydraulic conductivity can be more effective in retaining bacteria (Lo et al. 2002). Petersen et al. (2004) found greater accumulation of bromide in tilled than untilled soils, suggesting there was more bypass flow without ploughing. Water flow through the soil matrix will maximize filtration of bacteria (McMurray et al. 1998), leading to a dominance of filtration in the 0- to 10-cm layer. Any water carrying bacteria past the surface soil is then preferentially transported to deeper layers, a situation found in this study.

According to the literature, whether bacteria are transported through the soil matrix or through macropores will depend on the initial soil water content, and the amount and intensity of the input solution. Infiltration is not an instantaneous process and any local surface ponding could lead to runoff into cracks or into root channels. In light of this, it could be stated that as the solution input rate was greater than the infiltration rate for the given soil, excess water could circumvent the soil matrix and penetrate macropores, thus leading to all three treatments displaying transport similar to a wet or actively wetting state. This would account for the filtration and deposition of *E. coli* and Cl^- due to the wide pore size distribution conducting the solution, as well as the flow in organized subsurface pathways as evidenced by the similar contributing areas for all three treatments. High initial water content and high input flux density have been implicated in the enhanced transport of bacteria (Vinten et al. 2002). Generally, a high intensity rainfall, as well as an initially wet soil will transport a high proportion

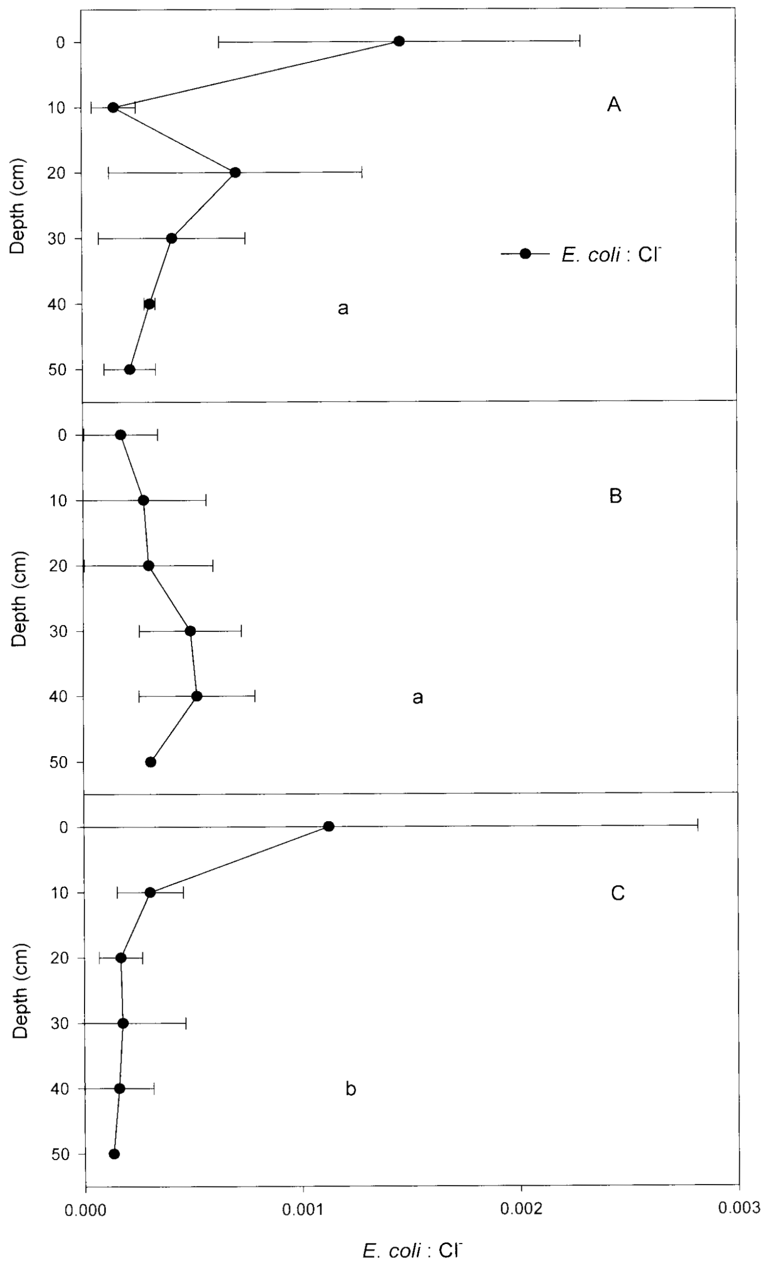

Fig. 4. Concentration ratio of *E. coli* to Cl^- relative to input ratios with respect to depth for Dry State (A), Actively Wetting State (B) and Wet State (C) treatments. Lower case letters indicate statistically significant differences at $P < 0.05$. The measurements were taken at discrete depths, and 0 indicates a surface measurement.

Table 3. *E. coli* repeated measures ANOVA table

Source	df (num/ den)[z]	*F* value	*P* > *F*
Between Treatment	2	0.89	0.4592
Wilk's λ (Depth)	4/3	3.09	0.1904
Wilk's λ (Depth × Treatment)	8/6	2.00	0.2062
Mean linear	1	6.72	0.0411*
Treatment linear	2	0.86	0.4707

[z]Indicates numerator degrees of freedom and denominator degrees of freedom.
*Indicates significance at $P < 0.05$.

Table 4. Cl^- repeated measures ANOVA table

Source	df (num/ den)[z]	*F* value	*P* > *F*
Between Treatment	2	0.22	0.8078
Wilk's λ (Depth)	4/3	7.37	0.0663
Wilk's λ (Depth × Treatment)	8/6	0.98	0.5252
Mean linear	1	41.20	0.0007*
Treatment linear	2	0.27	0.7734

[z]Indicates numerator degrees of freedom and denominator degrees of freedom.
*Indicates significance at $P < 0.05$.

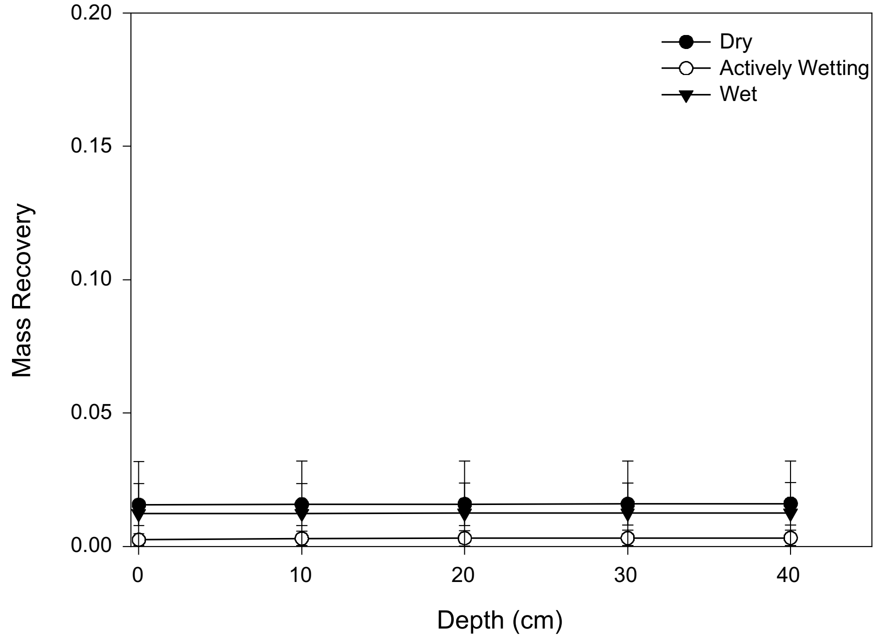

Fig. 5. Normalized cumulative infiltration trend (ratio of measured to total application) of *E. coli* with respect to depth.

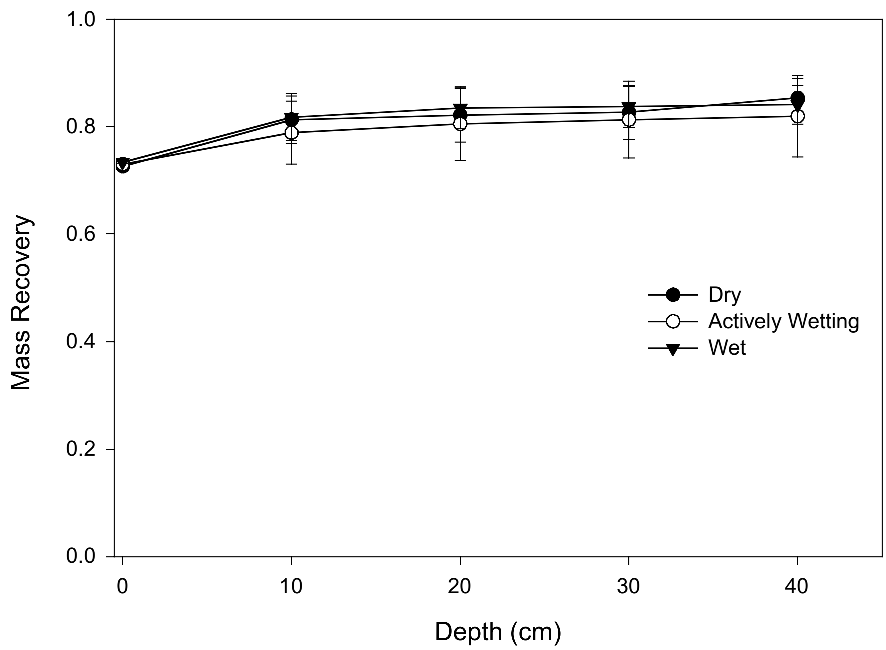

Fig. 6. Normalized cumulative infiltration trend (ratio of measured to total application) of Cl^- with respect to depth (%).

of bacteria to depth and possibly to groundwater (Unc and Goss 2003; Natsch et al. 1996). It is likely that there would have been different results in the present study had rainfall application occurred over a more prolonged period at a lower flux density. However, the input rates used are representative of convective storm events in semi-arid environments and are of the same magnitude (6.25×10^{-5} m s^{-1}) as a 25-yr return period storm for this area (Gray 1973). In the study area, rainfall events of between 1 to 2.5 cm take place about once per month during the summer (Environment Canada 2006).

It may be concluded that the mechanisms controlling transport in this study were similar for all three treatments. From the results obtained it is possible that the input solution formed a perched water table in the plough layer where the surface of the soil controlled transport to depth.

Evidence of a perched water table can be seen in the conducting areas, which are small and organized at depths below 10 cm while the surrounding soil remains un-stained. This saturated layer would have fed the preferential flow at successive depths and whatever water was released past the surface was transported along organized, conducive pathways. This result is similar to that of Weiler and Naef (2003), who found that macropore flow is initiated either at the soil surface, or from a saturated layer. As suggested in the literature (Smith et al. 1985; Abu-Ashour et al. 1998), the majority of *E. coli* and tracers that were applied are subject to matrix flow in small tortuous pores, and thus are trapped and filtered out of solution. However, some amount of the mass applied will enter pathways conducive to transport, such as connected macropores or root channels. Clear evidence for preferential transport is given in the fact that even though only 1.9 cm of water was applied, the soil was stained to a depth of 50 cm.

The input solution that was allowed past the soil surface entered organized channels connected to deeper depths. During excavation and sampling of the experimental plots in the present study, continuous cracks open to the surface did not exist. It was originally hypothesized that open desiccation cracks could transport a large proportion of the input solution to a depth that would bypass the surface entirely. This was not the case as desiccation cracks had swollen shut when the solution was applied. As was the case for the results in Aislabie et al. (2001), it was a structured soil and macropore channels that were the dominant transport mechanisms. The present study is similar to the heavy clay soil in Nobles et al. (2003), where *E. coli* and tracers were not only transported along subsurface root channels, but were organized along areas of soil corresponding to the seed row, allowing for preferential transport beyond the surface.

Clay soils have small matrix conductivities that are easily overwhelmed by even a small intensity rainfall. The rainfall in the present study, though representative of the area, was intense enough to create a saturated zone with the only outlet being along subsurface root channels. Despite the presence of a plough layer with a discontinuous pore network, the macroporosity of the study field in the 0–10 cm layer was 0.073 $m^3\ m^{-3}$ and K_s as measured by tension infiltrometer, was $3.01 \times 10^{-5}\ ms^{-1}$. Since root channels are important in the transport of bacteria, this would allow for transport of *E. coli* when the flux and volume of the input solution were sufficient to overcome infiltration rates, thus allowing larger pores to become active. This served to minimize the effect of the antecedent preferred wetting state of the soil.

From the results of this study certain findings become apparent: of the portion of *E. coli* and Cl^- that were applied, the majority were retained in the surface layer of the soil; the flux density of the input solution was sufficient to override antecedent soil wetting state, leading to organized flow pathways and negating any statistically significant treatment effects on transport. All of these results are indicative of a wet preferred state (Grayson et al. 1997).

In this study, we chose the *E. coli* strain because it was a well characterized strain that is native to beef cattle. For practical enumeration and detection purposes, the strain had to be modified with the gfp and antibiotic resistance. The strain did not compete well in a natural environment, but the focus of the research was the transport of *E. coli* and not the persistence and recovery. We feel that the strains were sufficiently similar to report that the transport that was measured is indicative of an organism that has not been modified.

CONCLUSIONS

The results presented here raise the question of timing in manure spreading. It was suggested by McGechan and Vinten (2004) that livestock grazing only be allowed during dry seasons and be discontinued when wet soil conditions prevail. The present study has shown that in semi-arid agricultural areas with clay loam or clay soil texture, the risk of *E. coli* transport through the soil profile is equally prevalent regardless of preferred state. If there is an intense thunderstorm after application of manure, *E. coli* can be transported along preferential flow paths to at least 50 cm depth. The *E. coli* present in manure can survive and be transported even in a clay soil, widely thought to enhance filtration, for as long as 16 d after application (Saini et al. 2003). It is suggested that in order to err on the side of caution, measures such as composting should be taken to minimize the *E. coli* population, and areas with shallow water tables should be avoided altogether.

Abu-Ashour, J., Joy, D. M., Lee, H., Whitely, H. R. and Zelin, S. 1998. Transport of microorganisms through soil. Water Air Soil Pollut. **75**: 141–158.

Aislabie, J., Smith, J. J., Fraser, R. and McLeod, M. 2001. Leaching of bacterial indicator of faecal contamination through four New Zealand soils. Aust. J. Soil Res. **39**: 1397–1406.

Artz, R. R. E., Townend, J., Brown, K., Towers, W. and Killham, K. 2005. Soil macropores and compaction control the leaching potential of *Escherichia coli* 0157:H7. Environ. Microbiol. **7**: 241–248.

Celico, F., Varcamonti, M., Guida, M. and Naclerio, G. 2004. Influence of precipitation and soil on transport of fecal *Enterococci* in fractured limestone aquifers. Appl. Environ. Microbiol. **70**: 2843–2847.

Environment Canada. 2006. Canadian climate normals 1971-2–2000 [Online]. Available at: http://www.climate.weatheroffice.ec.gc.ca/climate_normals/results_e.html?Province=ALL&StationName=rosetown&SearchType=BeginsWith&LocateBy=Province&Proximity=25&ProximityFrom=City&StationNumber=&IDType=MSC&CityName=&ParkName=&LatitudeDegrees=&LatitudeMinutes=&LongitudeDegrees=&LongitudeMinutes=&NormalsClass=A&SelNormals=&StnId=3250&&autofwd=1 [2006 Jan. 03]. Environment Canada, Ottawa, ON.

Flury, M. and Wai, N. N. 2003. Dyes as tracers for vadose zone hydrology. Reviews of Geophysics. **41**. doi:10.1029/2001RG000109.

Gagliardi, J. V. and Karns, J. S. 2000. Leaching of *Escherichia coli* 0157:H7 in diverse soils under various agricultural management practices. Appl. Environ. Microbiol. **66**: 87–883.

Gray, D. M. 1973. Handbook on the principles of hydrology. Water information center. Port Washington, New York, NY.

Grayson, R. B., Western, A. W., Chiew, F. H. S. and Blöschl, G. 1997. Preferred states in spatial soil moisture patterns: Local and nonlocal controls. Water Resour. Res. **33**: 2897–2908.

Jones, D. L. 1999. Potential health risks associated with the persistence of *Escherichia coli* 0157 in agricultural environments. Soil Use Manage. **15**: 76–83.

Kulli B., Gysi, M. and Fluhler, H. 2003. Visualizing soil compaction based on flow pattern analysis. Soil Tillage Res. **70**: 29–40.

Kung, K.-J. S., Steenhuis, T. S., Kladivko, E. J., Gish, T. J., Bubenzer, G. and Helling, C. S. 2000a. Impact of preferential flow on the transport of adsorbing and non-adsorbing tracers. Soil Sci. Soc. Am. J. **64**: 1290–1296.

Kung, K.-J. S., Kladivko, E. J., Gish, T. J., Steenhuis, T. S., Bubenzer, G. and Helling, C. S. 2000b. Quantifying preferential flow by breakthrough of sequentially applied tracers: silt loam soil. Soil Sci. Soc. Am. J. **64**: 1296–1304.

Lo, K. W., Jin, Y.-C. and Viraraghavan, T. 2002. Transport of bacteria in heterogeneous media under leaching conditions. J. Environ. Eng. **1**: 383–395.

McGechan, M. B. and Vinten, A. J. A. 2004. Simulating transport of *E. coli* derived from faeces of grazing livestock using the MACRO model. Soil Use Manage. **20**: 195–202.

McMurray, S. W., Coyne, M. S. and Perfect, E. 1998. Fecal coliform transport through intact soil blocks amended with poultry manure. J. Environ. Qual. **27**: 86–92.

Meredith, M. P. and Stehman, S. V. 1991. Repeated measures experiments in forestry: focus on analysis of response curves. Can. J. For. Res. **21**: 957–965.

Natsch, A., Keel, C., Troxler, J., Zala, M., von Albertini, N. and Défago, G. 1996. Importance of preferential flow and soil management in vertical transport of a biocontrol strain of *Pseudomonas fluorescens* in structured field soil. Appl. Environ. Microbiol. **62**: 33–40.

Nobles, M. M., Wilding, L. P. and McInnes, K. J. 2003. Soil structural interfaces in some Texas Vertisols and their impact on solute transport. Catena **54**: 477–493.

Paton, J. C. and Paton, A. W. 1998. Pathogenesis and diagnosis of shiga toxin-producing *Escherichia coli* infections. Clin. Microbiol. Rev. **11**: 450–479.

Petersen, C. T., Hansen, S., Jensen, H. E., Holm, J. and Koch, B. 2004. Movement of suspended matter and a bromide tracer to field drains in tilled and untilled soil. Soil Use Manage. **20**: 271–280.

Saini, R., Halverson, L. J. and Lorimor, J. C. 2003. Rainfall timing and frequency influence of leaching of *Escherichia coli* RS2G through soil following manure application. J. Environ. Qual. **32**: 1865–1872.

Smith, M. S., Thomas, G. W., White, R. E. and Ritonga, D. 1985. Transport of *Escherichia coli* through intact and disturbed soil columns. J. Environ. Qual. **14**: 87–91.

Suarez A., Güttler, A., Strätz, M., Staendner, L. H., Timmis, K. N. and Guzmán, C. A. 1997. Green fluorescent protein-based reporter systems for genetic analysis of bacteria including monocopy applications. Gene **196** **(1-2)**: 69–74.

Topp, G. C., Davis, J. L. and Annan, A. P. 1980. Electromagnetic determination of soil-water content: Measurement in coaxial transmission line. Water Resour. Res. **16**: 574–582.

Topp, E., Welsh, M., Tien, Y.-C., Dang, A., Lazarovits, G., Conn, K. and Zhu, H. 2003. Strain-dependent variability in growth and survival of *Escherichia coli* in agricultural soil. FEMS Microbiol. Ecol. **44**: 303–308.

Unc, A. and Goss, M. J. 2003. Movement of faecal bacteria through the vadose zone. Water Air Soil Pollut. **149**: 327–337.

Unc, A. and Goss, M. J. 2004. Transport of bacteria from manure and protection of water resources. Appl. Soil Ecol. **25**: 1–18.

Varian. 1985. Flame atomic absorption spectroscopy analytical methods. Publication No. 85-100009-00. Victoria, Australia

Velde, B., Moreau, E. and Terribile, F. 1996. Pore networks in an Italian Vertisol: quantitative characterization by two dimensional image analysis. Geoderma. **72**: 271–283.

Vinten, A. J. A., Lewis, D. R., Fenlon, D. R., Leach, K. A., Howard, R., Svoboda, I. and Ogden, I. 2002. Fate of *Escherichia coli* and *Escherichia coli* 0157 in soils and drainage water flowing cattle slurry application at 3 sites in southern Scotland. Soil Use Manage. **18**: 223–231.

Weiler, M. and Flühler, H. 2004. Inferring flow types from dye patterns in macroporous soils. Geoderma **120**: 137–153.

Weiler, M. and Naef, F. 2003. An experimental tracer study of the role of macropores in infiltration in grassland soils. Hydrol. Process. **17**: 477–493.

Western, A. W., Grayson, R. B. and Blöschl, G. 2002. Scaling of soil moisture: A hydrologic perspective. Annu. Rev. Earth Planet. Sci. **30**:149–180.

Zehe, E. and Flühler, H. 2001. Preferential transport of isoproturon at a plot scale and a field scale tile-drained site. J. Hydrol. **247**: 100–115.

Zeleke, T.B. 2003**.** Surface soil physical properties after crop residue incorporation and injection of liquid swine effluent. M.Sc. thesis. Department of Soil Science. University of Saskatchewan. Saskatoon, SK.